

Ujjawal Pandey

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TECHNICAL SKILLS

Programming Languages:	Python, C++, SQL
Generative AI:	LLMs, Transformer Architecture, RAG Pipelines, Fine-tuning (LoRA/PEFT), LangChain, LanGraph, LlamaIndex, PageIndex(Vectorless), Hugging Face
Machine Learning:	PyTorch (nn.Module, Tensors), NumPy, Pandas, Scikit-learn, Deep Learning, NLP
Frontend & Backend:	React.js, FastAPI, Node.js, REST APIs, JWT Authentication
Databases & Tools:	MongoDB, PostgreSQL, Vector Embeddings, ChromaDB, Docker, AWS, Git, Linux

WORK EXPERIENCE

Summer Research Intern (On-Site) - SRM University AP	April 2025– September 2025
- Architected a context-aware RAG pipeline using LangChain and ChromaDB to automate question generation, implementing a personalized feedback loop that prioritizes topics from a student's past mistakes in subsequent assessments. - Researched an automated assessment framework integrating AI-driven OMR and secure backend protocols, analyzing the feasibility of tab-switch detection for low-bandwidth environments.	
Jeevika Services (Hybrid) - Janakpur, Nepal	October 2025- March 2026
- Assisted in developing a RAG-based chatbot by integrating LangChain to enable real-time retrieval of marketplace data, ensuring accurate responses based on live inventory. - Contributed to a price-sensitive recommendation engine that identifies optimal products for users, helping to automate the selection process based on real-time pricing and item specifications.	

PROJECTS

NLP Simplifier <i>React, FastAPI, HuggingFace (Trainer, Accelerate), Flan-T5, PyTorch</i>	
- Engineered a high-performance Seq2Seq text simplification model by performing full supervised fine-tuning of Flan-T5-Base on the WikiLarge dataset, utilizing Beam Search Decoding to successfully improve ROUGE-L (33 to 60) and BLEU (23 to 42). - Architected a memory-efficient training pipeline for 6GB VRAM hardware via HuggingFace Trainer and Accelerate, processing 250K data pairs in 8 hours through Gradient Accumulation, Gradient Checkpointing, AdamW, and Label Smoothing.	
AI Tutor <i>React, FastAPI, Ollama (Llama 3, DeepSeek Coder), Manim, Wav2Lip, Sarvam API</i>	
- Architected a multimodal AI tutoring platform deploying a dual-LLM Ollama environment. Leveraged Llama 3 for NLP and DeepSeek Coder for automated Manim scripting, rendering 30-second animated videos in under a minutes. - Engineered an interactive React frontend featuring a dynamic avatar, integrating the Sarvam API for natural text-to-speech narration and Wav2Lip for real-time, high-fidelity lip-syncing to deliver a seamless educational experience.	
Exam-Zone <i>React, Node.js, FastAPI, MongoDB, LangChain, ChromaDB, Groq/Ollama</i>	
- Engineered a production-grade RAG pipeline (LangChain, ChromaDB) to automatically generate contextually accurate exam questions from uploaded course materials, enabling educators to create full assessments in under 5 minutes. - Built a highly secure assessment architecture (Node.js, FastAPI, MongoDB) integrating real-time tab-switch detection for anti-cheat proctoring, alongside Groq/Ollama for low-latency semantic grading.	

EDUCATION

SRM University, AP	July 2023 – Present
Bachelor of Computer Science and Engineering	CGPA: 8.7/10
Coursework: Data Structure and Algorithm, DBMS, OOPS, OS, Computer Network, ML, Gen-AI, NLP	

ACHIEVEMENT

Code4change National Hacakthon	
1st Place, (KL University): Developed an innovative AI-driven Automated Assessment System recognized for excellence.	
CodeSpark National Hacakthon	
Winner, (KBN College) : Outperformed 500+ teams by developing an AI video generator for instant educational solutions.	

Certificate

Pytorch for AI <i>Infosys Springboard</i>
Advanced Generative AI for Developers <i>Google Cloud Skills Boost</i>